

The S-Curve After the Shock: Machine Learning Evidence on Mortgage Prepayment in the Lock-In Era

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Abstract

We estimate loan-level mortgage prepayment hazard models—logistic regression, gradient-boosted trees, and a deep neural network in the spirit of [Sadhvani et al. \(2021\)](#)—on 18.5 million Fannie Mae loans observed monthly between 2018 and 2025, and use them to characterize how the prepayment function changed after the 2022 interest-rate shock. We report three findings. First, the regime break was costly in both directions: a model estimated through 2021 over-predicts aggregate speeds by approximately 9 CPR points at the onset of the shock, yet annual re-estimation *worsens* 2023 forecasts relative to the frozen model (3.8 versus 2.3 CPR points of UPB-weighted cohort error), because re-estimation on the brief transition period imports an upwardly biased speed level. Re-estimation dominates only from 2025 onward. Second, partial-dependence analysis of the learned prepayment function on a common reference population shows that the lock-in era did not flatten the refinancing S-curve but rotated it: the refinancing response above the at-the-money point is substantially steeper than in the pre-pandemic regime (30.5 versus 13.2 CPR at +150 basis points of incentive), while deep-discount turnover operates primarily through the prevailing-rate level. Third, applying the estimated model to the December 2025 cross-section, the 2023–24 high-coupon universe exhibits pronounced negative convexity: projected speeds for 7.0% coupons rise from approximately 25 CPR at unchanged rates to above 50 CPR under a 150 basis point rally. We document three data-quality pitfalls in the public dataset that mechanically inflate out-of-sample classification metrics if untreated, and we release all code and artifacts for replication.

Keywords: mortgage prepayment; MBS; machine learning; lock-in effect; gradient boosting; interpretability.

1 Introduction

Between March 2022 and October 2023 the 30-year fixed mortgage rate in the United States rose from below 4% to nearly 8%, the fastest repricing of the mortgage market in four decades.

*This paper is a personal research project based exclusively on publicly available data. The views expressed are the author’s own and do not represent those of any employer. Code and replication materials are available in the accompanying public repository. Not investment advice.

Aggregate conditional prepayment rates (CPR) on Fannie Mae collateral fell from above 35 to below 5, and by the end of 2023 nearly the entire agency universe was out of the money. The episode placed prepayment models—the central pricing input for a \$9 trillion market—outside the support of their training data. The practitioner record of the aftermath is extensive: vendor models missed monthly aggregate prints by 10–25% through 2023–2025, disagreed on the sign of convexity below par, and generated relative-value signals that inverted against realized returns (Yu, 2023; Katz and Li, 2024).

A parallel academic literature has quantified the underlying mechanism—rate lock-in—from the housing side. Batzer et al. (2024) estimate that each percentage point by which prevailing rates exceed a borrower’s contract rate reduces the probability of sale by 18.1%; Fonseca and Liu (2024) find a 9–16% reduction in moving probability per percentage point of lock-in; Liebersohn and Rothstein (2025) attribute a 16% decline in the mobility of mortgaged households to the mechanism. This literature measures the turnover component of prepayment, but does not address the model-side question that concerns fixed-income practitioners: *what exactly changed in the prepayment function, and how should data-driven models be managed across such a break?*

This paper provides, to our knowledge, the first public model-side treatment of the 2022 regime break estimated on open data. We construct a monthly discrete-time hazard panel from the Fannie Mae Single-Family Loan Performance dataset (18.5 million loans, 2018–2025), estimate three classes of prepayment model of increasing flexibility under a strict walk-forward protocol, and evaluate them at two levels: loan-level discrimination and—the economically relevant metric—UPB-weighted cohort-level CPR error against full-population actuals.

Our contributions are threefold. First, we quantify the *re-estimation trap*: while the pre-shock model fails immediately and visibly (Section 5.2), naive annual re-estimation underperforms the frozen legacy model for roughly two years after the break, because parameters fitted to the brief transition sample import a biased level. This finding has direct implications for model governance and is consistent with, but sharper than, the documented whipsaw of commercial models. Second, using per-regime estimation and partial-dependence analysis on a fixed reference population (Section 5.3), we decompose the apparent post-2022 flattening of the S-curve into composition and behavior. Conditional on composition, the refinancing limb of the lock-in-era S-curve is *steeper* than its pre-pandemic counterpart—the refinancing elasticity through the cusp rises from 2.2 to 12.6 CPR points per 100 basis points—while depressed deep-discount turnover is absorbed by the prevailing-rate level rather than by a decayed incentive response. This decomposition speaks directly to a live practitioner debate (Landy, 2024; Katz and Li, 2024). Third, we translate the estimates into a forward-looking statement about the current coupon stack (Section 5.4): as of December 2025, model-implied speeds for the 2023–24 vintage 6.5–7.0% coupons rise steeply under modest rallies, tripling for 6.5s between unchanged rates and a 150 basis point rally.

A secondary, methodological contribution concerns data quality. Three fields in the public dataset are blanked on a loan’s terminal record, so that—if used contemporaneously—their missingness perfectly reveals the prepayment event. Untreated, these produce out-of-sample AUCs of 0.9999 and would invalidate any study built on them. Section 3.4 documents the mechanism and the corrections, which we regression-test in the released code.

2 Related literature

Machine learning for mortgage risk. [Sadhvani et al. \(2021\)](#) estimate multi-state transition models on 120 million U.S. mortgages with deep neural networks and document large out-of-sample gains over logistic benchmarks, with prepayment exhibiting the strongest nonlinearities. [Zhang et al. \(2019\)](#) deploy neural networks for agency MBS prepayment at pool level and interpret the fitted model through implied S-curves, burnout, and seasonality; [Schultz and Fabozzi \(2021\)](#) make the case for gradient-boosted trees at loan level. [Blumenstock et al. \(2022\)](#) find machine-learning survival models dominate statistical benchmarks for competing default and prepayment risks; [Ojha and Lee \(2021\)](#) compare conventional and deep learners on Fannie Mae data through the financial crisis. Relative to this literature, our contribution is not architectural novelty but the regime question: all published samples of which we are aware end before or shortly after the 2022 break, and none addresses model management across it.

The lock-in effect. [Batzer et al. \(2024\)](#), [Fonseca and Liu \(2024\)](#), and [Liebersohn and Rothstein \(2025\)](#) establish the magnitude of rate lock-in for housing turnover, labor mobility, and welfare. We use their elasticities as external anchors for the turnover-suppression channel recovered by our models.

Practitioner record. [Yu \(2023\)](#) frames post-2022 prepayment modeling as an out-of-sample problem; monthly commentary by Santander US Capital Markets documents production model errors of 10–25% through 2023–25 and argues that apparent S-curve flattening reflects spread-at-origination (SATO) composition ([Landy, 2024](#)); [Katz and Li \(2024\)](#) document model-implied relative value inverting against realized returns in 2024 and, in early 2026, pool-level S-curves exceeding pandemic-era steepness. Our per-regime, composition-controlled S-curves reconcile these observations within a single estimated model.

3 Data

3.1 Loan-level performance data

We use the Fannie Mae Single-Family Loan Performance dataset, acquisition quarters 2018Q1 through 2025Q4: 18,468,917 fixed-rate loans and their complete monthly servicing records through December 2025, parsed from the raw 113-field pipe-delimited files. The window brackets three regimes: pre-pandemic (2018–19), the pandemic refinancing wave (2020–21), and the lock-in era (2022–25). Voluntary prepayment is identified by zero-balance code 01 (payoff); credit events and repurchases are treated as censoring.

3.2 Cohort-level ground truth

From the full population—prior to any sampling—we compute actual cohort-level single monthly mortality and CPR by vintage year and coupon (rounded to the nearest 0.5), where SMM is voluntarily prepaid UPB over beginning-of-month UPB. These series serve as the evaluation target for all models and align with public anchors: 2023-vintage 7.0 coupons print 5.4 CPR

in June 2024, peak at 30.3 in the October 2024 refinancing episode, and revert to 9.1 by December (Federal Housing Finance Agency, 2024); 2021-vintage 2.5 coupons remain at 2–4 CPR throughout 2023–25; 2018-vintage 4.5 coupons run at 41–57 CPR through the pandemic wave (Haughwout et al., 2023).

3.3 Estimation panel

A deterministic, hash-based stratified sample (vintage year $\times$ note-rate bucket, with inverse-probability loan weights) of 1,201,243 loans yields a monthly hazard panel of 2,532,987 loan-month observations containing all 428,793 voluntary prepayment events and a 5% importance-weighted subsample of non-event months. Weights are carried through estimation and aggregation, so fitted hazards are on the monthly SMM scale and cohort aggregates are population-representative.

Covariates comprise: the refinancing incentive (note rate minus the Freddie Mac PMMS 30-year survey rate); spread at origination (SATO); burnout (cumulative positive incentive over prior months); mark-to-market LTV (original LTV revalued with state-level FHFA house-price indices); loan age; original balance; credit score; DTI; original LTV; calendar month; the prevailing mortgage-rate level; and categorical origination attributes (channel, purpose, property type, occupancy, state, borrower count, first-time-buyer flag) plus lagged delinquency status and modification flag.

3.4 Removal-record measurement pitfalls

Three covariates initially produced out-of-sample AUCs of 0.9999—a value that should be read as diagnostic of leakage rather than of predictive skill. The mechanism is systematic: the dataset blanks monthly-reported fields on a loan’s terminal record. Consequently (i) mark-to-market LTV computed on the end-of-period balance equals exactly zero if and only if the loan prepaid that month; (ii) the modification flag and delinquency status are missing on removal records, so their missingness indicators reveal the event; and (iii) the reported loan-age field is likewise blanked. All state-dependent inputs are therefore measured as of the *beginning* of the month—balances lagged (the factor-date convention), status fields lagged one month, and loan age computed from calendar arithmetic rather than read from the file. Each correction carries a regression test in the released code. Post-correction discrimination falls to the 0.59–0.75 range, which we regard as the honest difficulty of monthly voluntary-prepayment classification.

4 Methodology

4.1 Hazard framework

Let $y_{it} \in \{0, 1\}$ indicate voluntary prepayment of loan i in month t . We model the discrete-time hazard $h_{it} = \Pr(y_{it} = 1 \mid x_{it})$, where x_{it} contains only information available at the beginning of month t . Cohort-level speeds aggregate fitted hazards with combined sampling and balance

weights: for cohort c in month t ,

$$\widehat{\text{SMM}}_{ct} = \frac{\sum_{i \in c} w_{it} b_{it} \hat{h}_{it}}{\sum_{i \in c} w_{it} b_{it}}, \quad \widehat{\text{CPR}}_{ct} = 1 - (1 - \widehat{\text{SMM}}_{ct})^{12}, \quad (1)$$

where w_{it} is the inverse-probability sampling weight and b_{it} the balance entering the month.

4.2 Estimators

We estimate three models of increasing flexibility on identical features and weights: (i) weighted logistic regression (median-imputed numerics, one-hot categoricals)—the linear benchmark; (ii) gradient-boosted trees (LightGBM; 63 leaves, 600 trees, learning rate 0.05, minimum 100 observations per leaf), with missing values handled natively; and (iii) a feed-forward neural network with categorical embeddings (three hidden layers, 128–128–64 ReLU units, dropout 0.1), trained with weighted cross-entropy, following the architecture spirit of [Sadhvani et al. \(2021\)](#). We deliberately avoid early stopping on trailing-window validation sets: in the presence of regime shifts, a trailing window is unrepresentative of both past and future, and we found it induces severe underfitting precisely at the breaks.

4.3 Evaluation protocol

All evaluation is strictly walk-forward: models are estimated on months up to December of year T and evaluated on the following twelve months, for $T = 2018, \dots, 2024$. We report loan-level weighted AUC, and UPB-weighted mean absolute cohort CPR error (in CPR points) against the full-population actuals of Section 3.2. We emphasize the latter: loan-level discrimination metrics have limited economic meaning for portfolio speeds, and—as Section 3.4 illustrates—are also the metrics most easily corrupted by leakage.

5 Results

5.1 Walk-forward performance

Table 1 reports the walk-forward record. Three observations. First, the gradient-boosted model dominates the logistic benchmark on cohort-level error in five of seven windows, decisively so in the lock-in era (1.1 versus 6.1 CPR points at the 2024 split). Second, both models miss the level of the 2020–21 refinancing wave—as did commercial models; the determinants of wave amplitude (origination capacity, appraisal waivers, media effects) are not spanned by loan-level covariates. Third, discrimination *declines* for all models in the lock-in era: with the entire universe out of the money, the residual variation in who prepays is close to idiosyncratic. The neural network, evaluated at three splits, is competitive on discrimination (AUC 0.75/0.65/0.64) but weaker than the trees on cohort error (8.8/4.6/1.5 versus 8.8/2.8/1.1 CPR points) and under-predicts the hazard level; at 2.5 million training rows this is consistent with the tabular-learning literature, and the original deep-learning advantage of [Sadhvani et al. \(2021\)](#) was demonstrated on three orders of magnitude more data. The gradient-boosted model is therefore the engine for the experiments that follow.

Training window ends	Loan-level AUC		Cohort CPR MAE (pts)	
	Logistic	GBM	Logistic	GBM
2018-12	0.66	0.71	11.6	6.4
2019-12	0.75	0.72	12.7	8.8
2020-12	0.72	0.74	8.6	14.6
2021-12	0.61	0.66	4.6	2.8
2022-12	0.52	0.59	2.6	3.8
2023-12	0.57	0.63	5.2	2.6
2024-12	0.59	0.68	6.1	1.1

Table 1: Walk-forward evaluation. Each row: estimation through December of the stated year, evaluation on the subsequent twelve months. Cohort CPR MAE is UPB-weighted across vintage $\times$ coupon cohorts against full-population actuals. The 2020–12 test window is the peak of the pandemic refinancing wave.

5.2 The regime break and the re-estimation trap

We compare a model frozen at December 2021 against annual re-estimation, both walked forward over 2022–25 (Figure 1). The frozen model’s failure is immediate: in January 2022 it predicts aggregate speeds of approximately 20 CPR against an actual print of 11, an error of nearly 9 CPR points, as it extrapolates refinancing-wave behavior into a repriced market. The subtler result concerns re-estimation. In 2023, the annually re-estimated model—now fitted through December 2022, including the transition—performs *worse* than the frozen model: 3.8 versus 2.3 CPR points of UPB-weighted error. Six months of partially transitioned data teach the model a speed level that is still too fast, while the frozen model’s errors, though structural, are stable and partially offsetting. Re-estimation overtakes the frozen benchmark only in 2024 (2.6 versus 2.1) and wins decisively in 2025 (1.1 versus 2.4). The governance implication: after a structural break, immediate recalibration on the transition sample can be value-destroying; approximately two years of new-regime data were required before re-estimation dominated. This is consistent with the documented 2023–25 whipsaw in commercial model performance (Katz and Li, 2024).

5.3 The learned S-curve, by regime

We estimate the gradient-boosted model separately on each regime—pre-pandemic (2018-01 to 2019-12), refinancing wave (2020-04 to 2021-12), and lock-in (2022-07 to 2025-12)—and trace each fitted model’s partial-dependence function of the refinancing incentive on a *common* reference population of 200,000 lock-in-era loan-months. Holding the population fixed ensures that differences between curves reflect learned behavior rather than composition. Figure 2 and Table 2 report the results.

Regime	CPR at -200bp	CPR at 0	CPR at $+150\text{bp}$	Elasticity (pts/100bp)
Pre-pandemic 2018–19	8.8	9.9	13.2	2.2
Refinancing wave 2020–21	20.4	23.9	35.1	7.5
Lock-in 2022–25	8.9	11.7	30.5	12.6

Table 2: S-curve landmarks by regime on the common reference population. Elasticity is the average slope between 0 and $+150$ basis points of incentive.

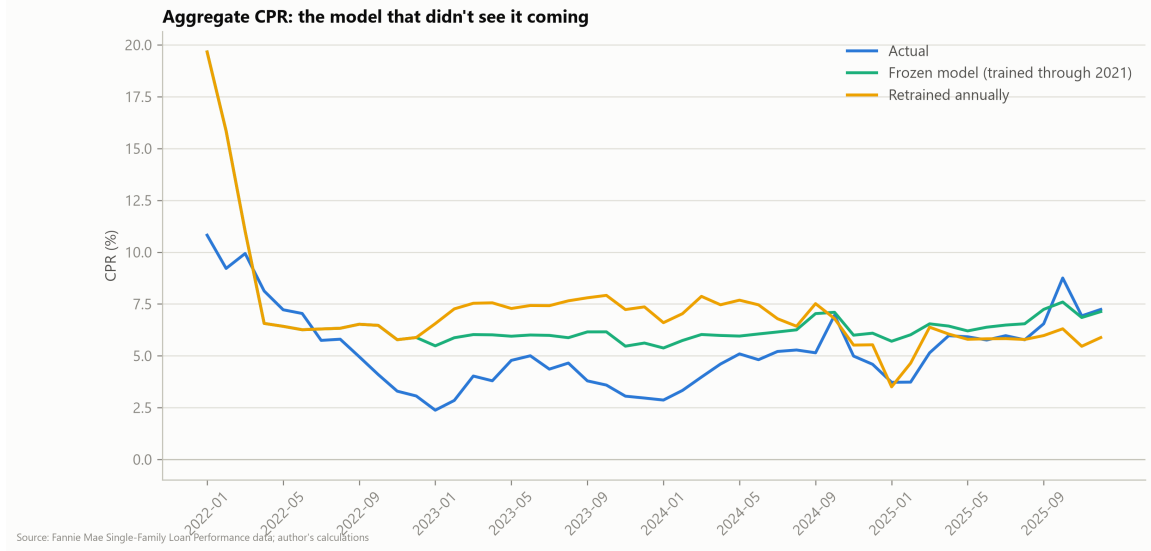


Figure 1: Aggregate UPB-weighted CPR, 2022–2025: actuals versus a model frozen at December 2021 and a model re-estimated each year-end. The two model series coincide during 2022 by construction. The frozen model over-predicts by ≈ 9 CPR points at the January 2022 onset; the re-estimated model over-predicts throughout 2023.

The central finding is a rotation, not a flattening. The refinancing limb of the lock-in-era S-curve is markedly steeper than its pre-pandemic counterpart: 30.5 versus 13.2 CPR at +150 basis points, an elasticity through the cusp of 12.6 versus 2.2 CPR points per 100 basis points. Borrowers who reach the money in the lock-in era refinance faster than any pre-2020 cohort would have—consistent with the 15-month average age at refinancing observed in the 2024 episode and with the efficiency of modern origination channels. At the deep-discount end, the lock-in model’s implied turnover on the common population is close to pre-pandemic levels; the observed 2–4 CPR prints of deeply out-of-the-money cohorts are reproduced through the prevailing-rate-level covariate—the lock-in channel of [Batzner et al. \(2024\)](#)—rather than through a degraded incentive response. SATO-conditioned partial dependence (replication materials) confirms the composition mechanism of [Landy \(2024\)](#): within every regime, higher-SATO tertiles trace steeper curves. The decomposition thus reconciles the composition view with the observation of [Katz and Li \(2024\)](#) that late-2025 realized S-curves exceed pandemic-era steepness: both are features of the same estimated function.

Attribution analysis (mean absolute SHAP values; replication materials) tells the same story at the level of drivers: pre-pandemic prepayment is dominated by burnout and seasoning; the wave by the incentive; and in the lock-in era all drivers compress, consistent with the decline in achievable discrimination.

5.4 Scenario read-through for the current coupon stack

Finally, we apply the full-sample model to the December 2025 cross-section of active 2023–24 vintage 6.0–7.0% coupons under parallel shifts of the mortgage rate, holding burnout and house prices fixed (a deliberately transparent one-month-hazard exercise, annualized). Table 3 reports projected speeds.

The 7.0% coupons are already fast at unchanged rates (≈ 25 CPR) and reach refinancing-

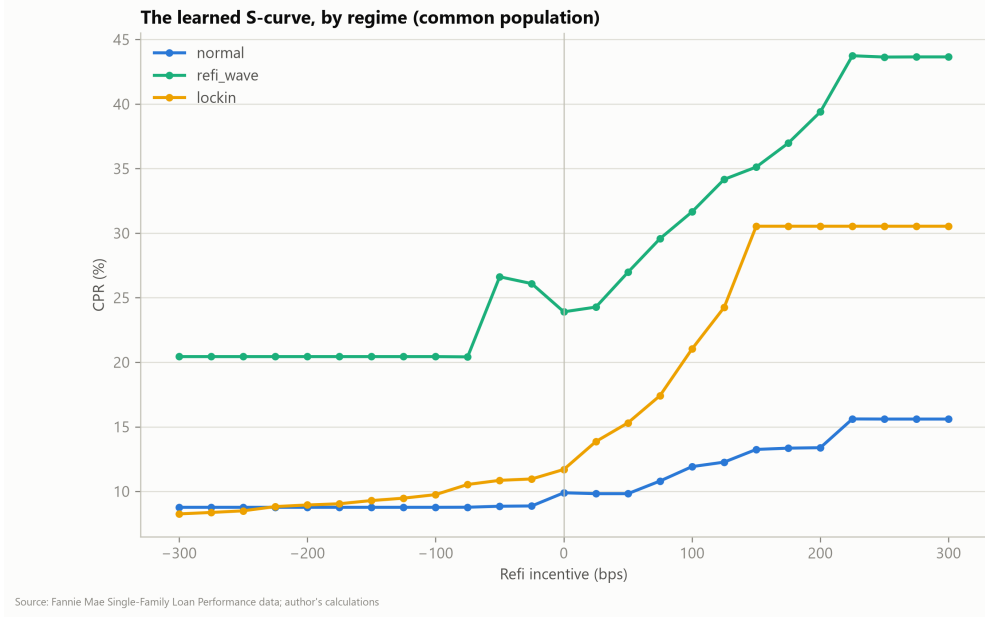


Figure 2: Model-implied S-curves (partial dependence of annualized speed on refinancing incentive) by estimation regime, evaluated on a common lock-in-era reference population.

Cohort	Unchanged	−50bp	−100bp	−150bp
2023 6.0s	6.9	12.9	20.7	25.9
2023 6.5s	13.1	22.5	29.1	36.0
2023 7.0s	25.4	34.4	43.7	52.0
2024 6.0s	6.5	12.1	19.4	26.1
2024 6.5s	13.3	23.0	32.4	42.1
2024 7.0s	25.5	35.5	43.7	52.2

Table 3: Model-projected CPR for the December 2025 cross-section of 2023–24 vintage cohorts under parallel mortgage-rate shifts.

wave speeds (above 50 CPR) under a 150 basis point rally; the 6.5s approximately triple over the same interval and constitute the cuspiest segment. This is the model-implied counterpart of the record negative convexity measured in the current-coupon market in late 2025 (Katz and Li, 2024), localized to specific vintage-coupon cohorts.

6 Limitations

Our sample is Fannie Mae collateral only; Ginnie Mae collateral—whose VA segment exhibited the fastest and most policy-sensitive speeds of the 2024–25 episodes—is outside the dataset. The pre-pandemic regime is observed for only two years (vintages 2018+). The scenario analysis is a one-month hazard annualized with burnout and house prices held fixed; it understates path dependence in a sustained rally and is not a valuation (OAS) model. The level of the 2020–21 wave is under-predicted by all specifications, ours and commercial alike, reflecting absent capacity and media-effect covariates. Partial-dependence curves are counterfactuals on a fixed population, not cohort forecasts. Finally, monthly PMMS timing and the publication lag of quarterly house-price indices introduce weeks-level measurement slack, disclosed here and in the

replication materials.

7 Conclusion

Using open data and commodity hardware, we provide the first public model-side account of the 2022 mortgage prepayment regime break. The prepayment function did not merely shift: its refinancing limb steepened while its turnover component became governed by the prevailing-rate level, and the transition itself created a two-year window in which naive model re-estimation was counterproductive. For practitioners, the results argue for explicit regime-awareness in model governance—freezing, blending, or regime-conditioning rather than reflexive recalibration—and for close attention to the steep embedded optionality of the 2023–24 high-coupon stack. For researchers, the removal-record leakage taxonomy of Section 3.4 is a cautionary catalogue for anyone estimating event models on the public agency datasets: an out-of-sample AUC of 0.9999 is not a discovery.

References

- Batzer, R., Coste, J., Doerner, W., and Seiler, M. (2024). The lock-in effect of rising mortgage rates. Staff Working Paper 24-03, Federal Housing Finance Agency.
- Blumenstock, G., Lessmann, S., and Seow, H.-V. (2022). Deep learning for survival and competing risk modelling. *Journal of the Operational Research Society*, 73(1):26–38.
- Federal Housing Finance Agency (2024). Prepayment monitoring reports (quarterly). FHFA, Division of Research and Statistics.
- Fonseca, J. and Liu, L. (2024). Mortgage lock-in, mobility, and labor reallocation. *Journal of Finance*, 79(6):3729–3772.
- Haughwout, A., Lee, D., Scally, J., and van der Klaauw, W. (2023). The great pandemic mortgage refinance boom. Federal Reserve Bank of New York, Liberty Street Economics.
- Katz, D. and Li, H. (2024). Models versus reality. TCW Securitized Products Spotlight.
- Landy, B. (2024). Don’t be fooled by flatter S-curves. Santander US Capital Markets, Portfolio Strategy.
- Liebersohn, J. and Rothstein, J. (2025). Household mobility and mortgage rate lock. *Journal of Financial Economics*, 164.
- Ojha, V. and Lee, K. (2021). Default analysis in mortgage risk with conventional and deep machine learning. *Digital Finance*, 3:249–271.
- Sadhwani, A., Giesecke, K., and Sirignano, J. (2021). Deep learning for mortgage risk. *Journal of Financial Econometrics*, 19(2):313–368.
- Schultz, G. and Fabozzi, F. J. (2021). Rise of the machines: Application of machine learning to mortgage prepayment modeling. *Journal of Fixed Income*, 31(3):6–19.

Yu, Y. (2023). Agency MBS in 2023: Uncharted territory. MSCI Research Insight.

Zhang, J., Zhao, Y., Zhang, H., Teng, F., Lin, S., and Li, J. (2019). Agency MBS prepayment model using neural networks. *Journal of Structured Finance*, 24(4):17–33.